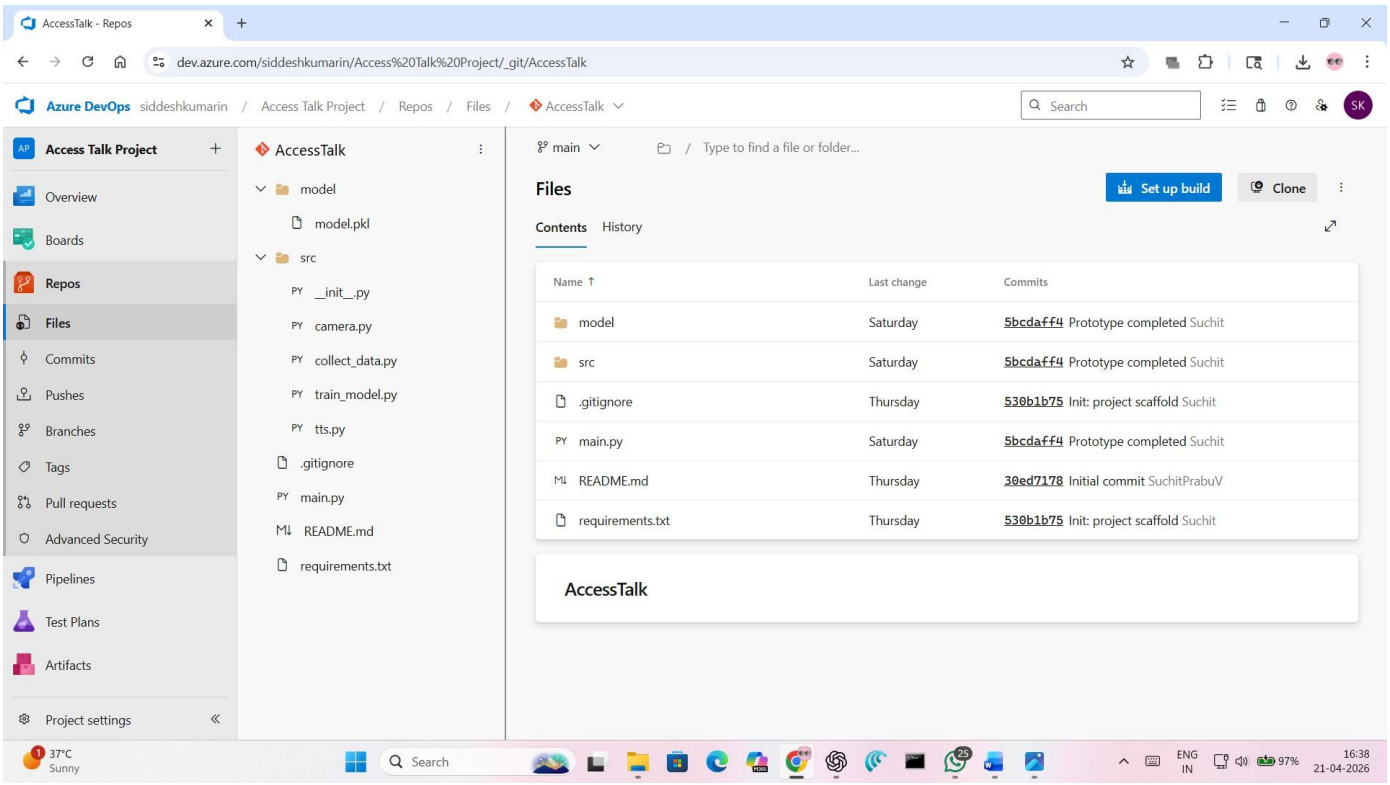
Experiment 9 : Implementation

Aim

To develop a **real-time Sign Language to Speech system (AccessTalk)** that uses a webcam to detect hand gestures, converts them into text using a machine learning model, and outputs the result as both on-screen text and speech without using any server or cloud.

Procedure

1. Create a project repository and setup folder structure (src, model, etc.).
2. Collect gesture data using collect_data.py by capturing images for each sign.
3. Train a machine learning model using train_model.py with MediaPipe landmarks and Random Forest algorithm.
4. Save the trained model as model.pkl in the model folder.
5. Develop camera pipeline (camera.py) to capture frames and extract hand landmarks.
6. Implement text-to-speech module (tts.py) to convert predicted text into speech.
7. Integrate all modules in main.py:
    - Capture video ,Extract landmarks ,Predict gesture ,Display text ,Generate speech
8. Run the application and test real-time gesture recognition using webcam.



Github Link : https://github.com/SiddeshKumarL/AccessTalk_Project.git